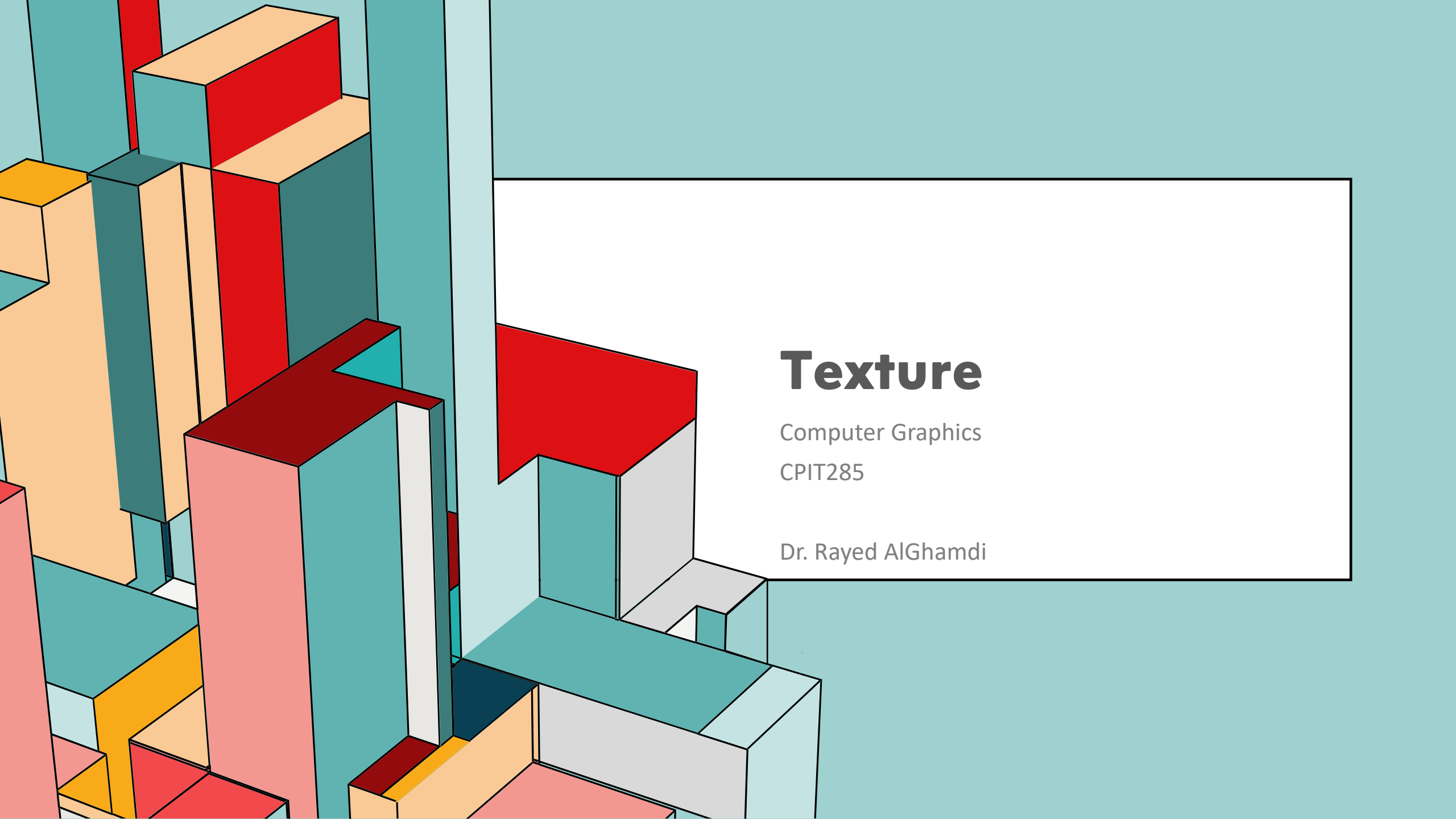




Texture

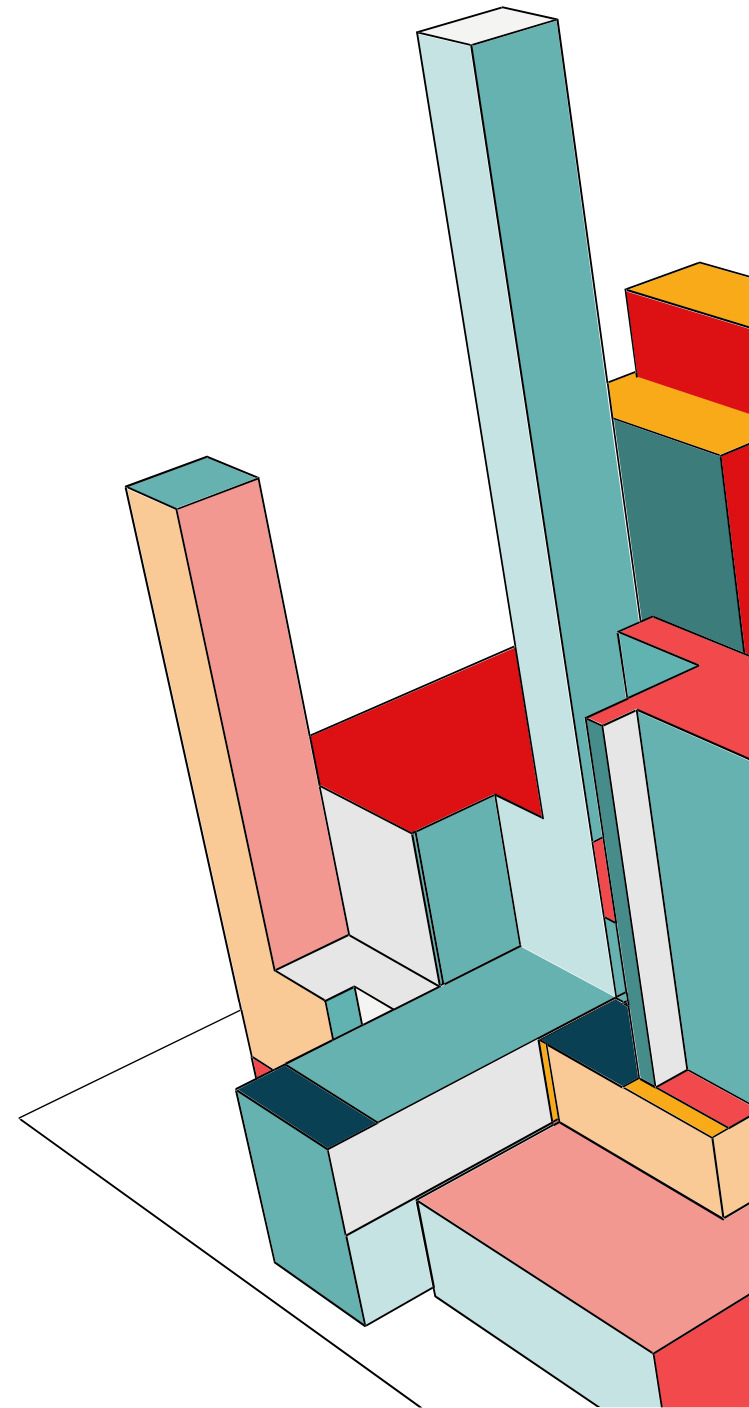
Computer Graphics

CPIT285

Dr. Rayed ALGhamdi

CONTENT

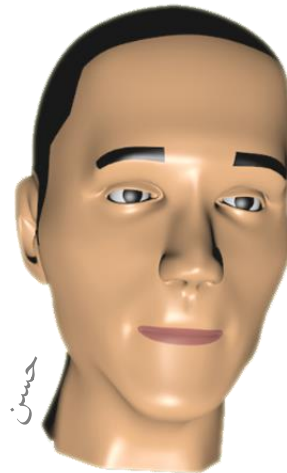
- Introduction
- UV Mapping & Unwrapping
- Photorealistic Texturing
- High Poly/Low Pply Baking
- OpenGL Texturing Functions



TEXTURING

Getting Started

- To what extent would coloring an object and creating lighting effects simulate realism?
- Think of your head model: you colored it, played with light effects... still not even close to your real head/skin!



TEXTURING

Getting Started

- How real skin, hair, eyes... can be simulated?

ARAB MAN- RIGGED 3D MODEL



ON SALE
\$50.25
~~\$67.00~~

MORE INFO

PRODUCT LICENSE

GEOMETRY:

About 50000 polygons for Body, Clothing, Hair, Mouth, Teeth and Tongue meshes.

RIGGED:

3DS Max Biped using SKIN modifier.

ANIMATED:

Ready to load eMOTION Mocap Data or any .bip motion files.

TEXTURES:

4K High quality Maps

(Diffuse, Specular, Bump, Yes - MoCap Data Included, Displacement, etc).

MATERIALS:

Mental Ray and V-Ray ready.

FORMAT:

Models are compatible with 3DS Max 2010 or above.

UNITS:

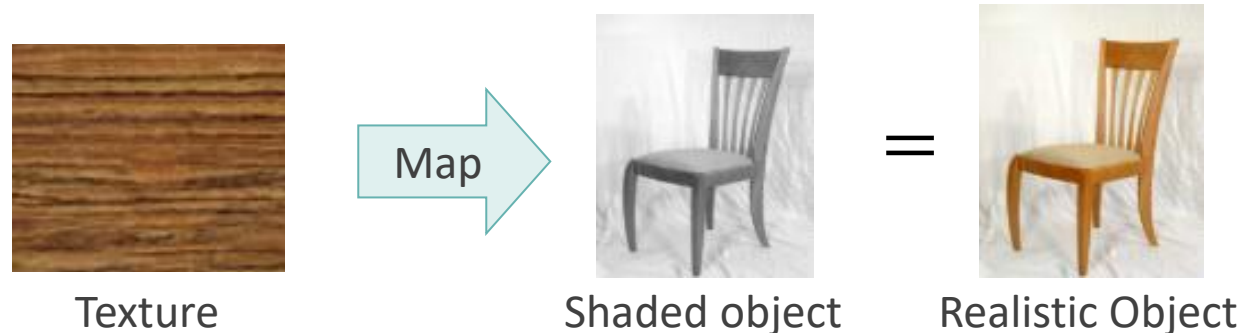
Centimeters and Inches.



TEXTURING

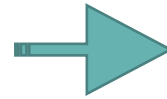
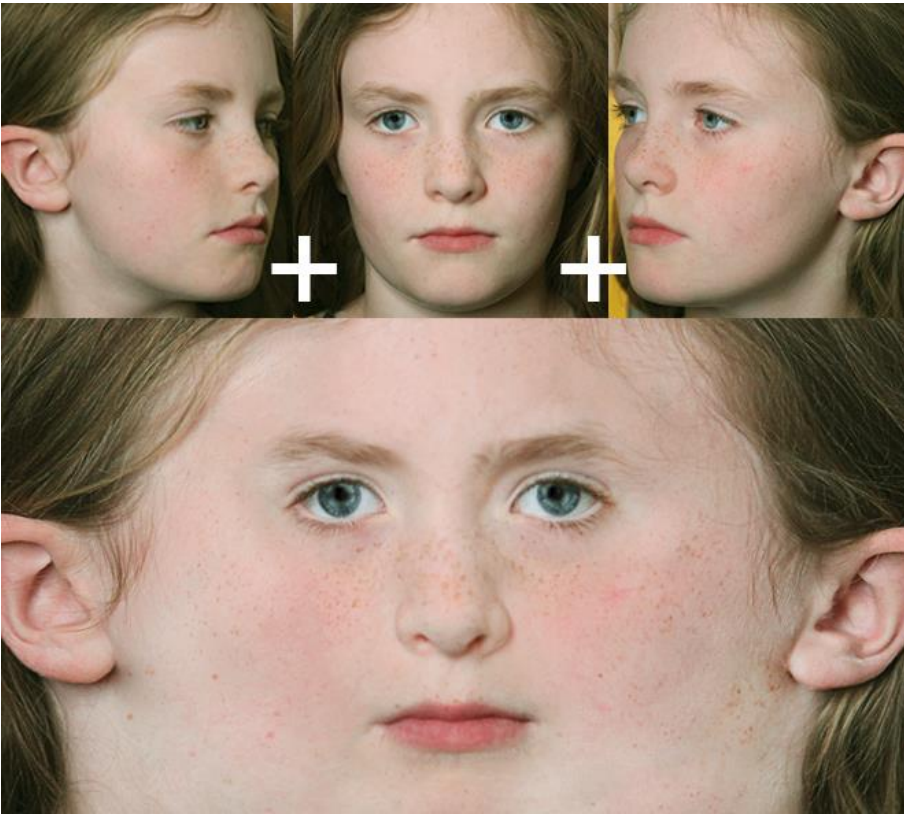
Getting Started

- To what extent would coloring an object and creating lighting effects simulate realism?
- The texture is an image in a 2-dimensional array of colors values.
- Texture mapping is a technique that adds detail to the surface of an object by mapping the pattern (texture) onto the geometrical description of the object.
- Purpose: To provide a realistic view of the 2D/3D object and the surrounding environment as a real-world environment.



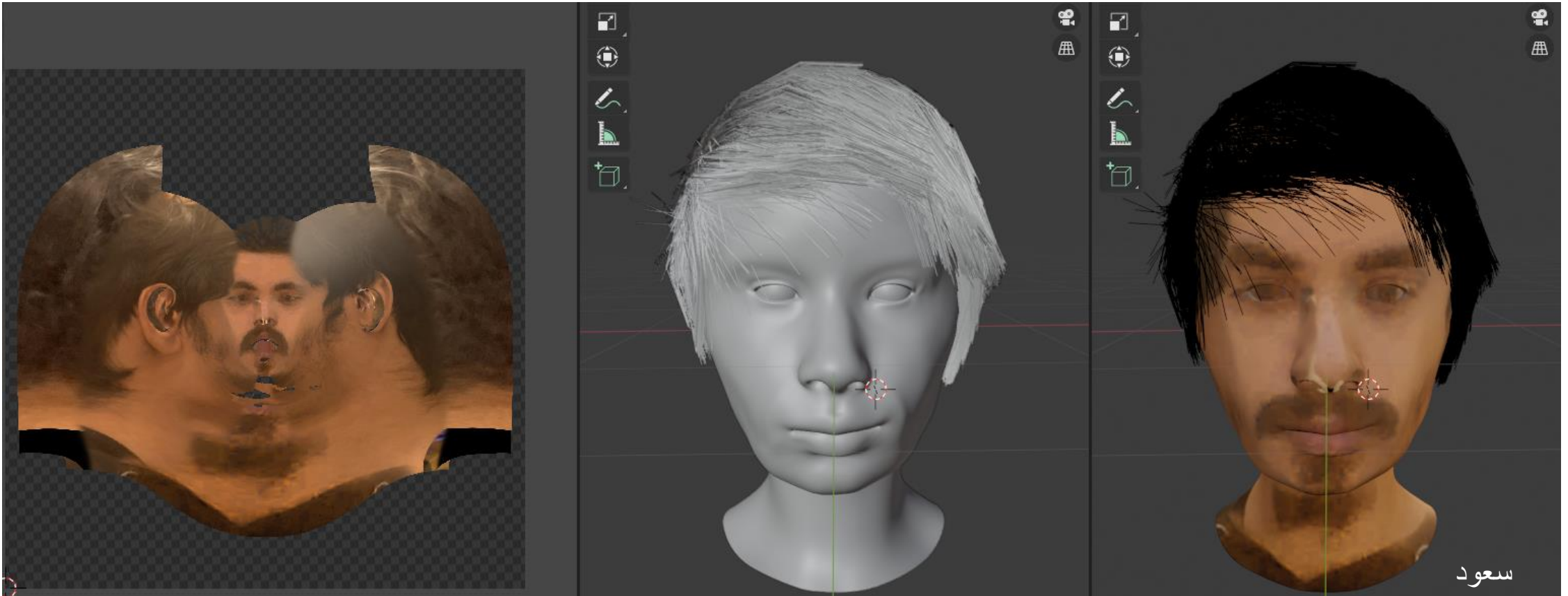
TEXTURING

Getting Started



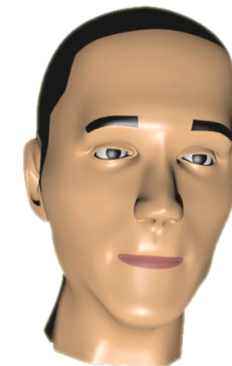
TEXTURING

Getting Started



TEXTURING

Getting Started

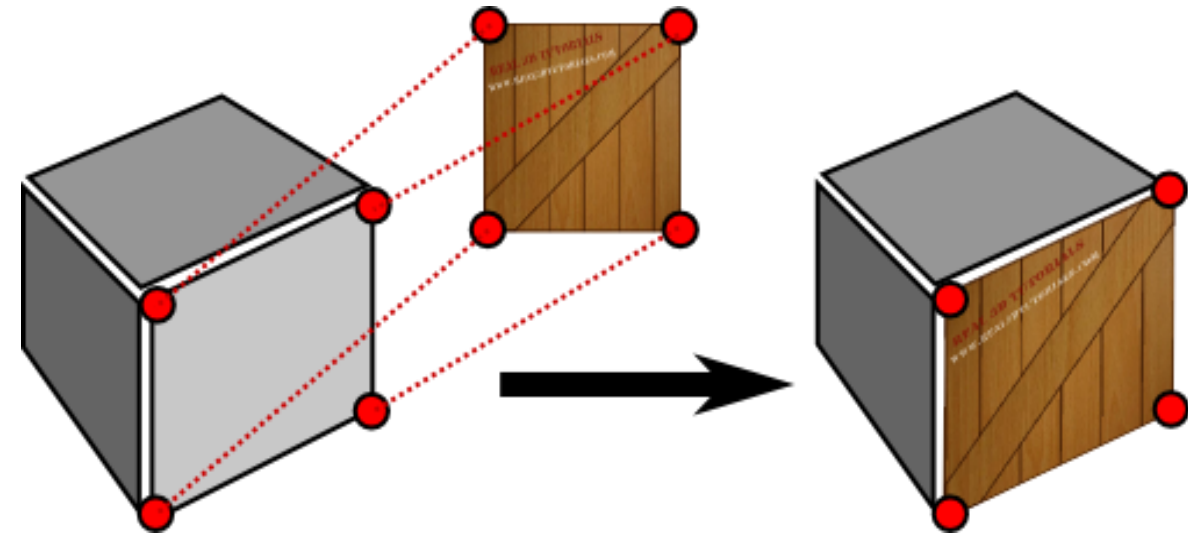


[Blender Skin Texturing and Shading tutorial](#)

TEXTURING

Texture Mapping

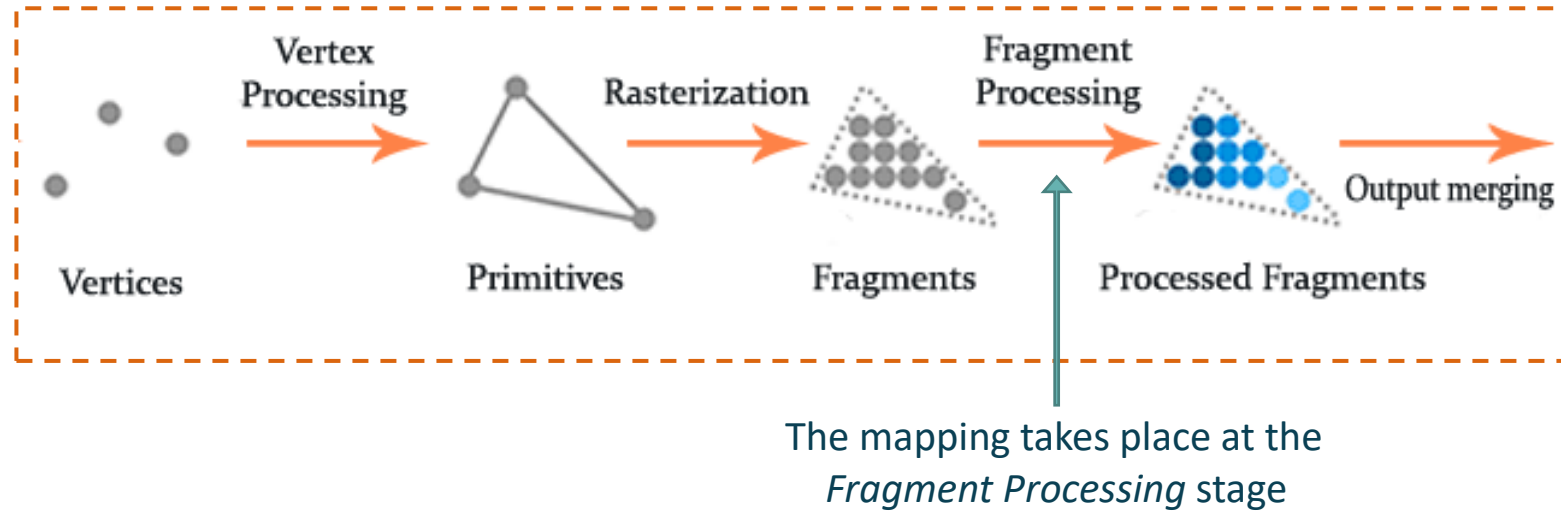
- Texture mapping is a common method for adding detail to an object.
- It is a technique used to paint an image onto a polygon.
- Texturing can be used for 2D or 3D objects.
- Texturing 2D objects is much easier compared to 3D objects.
- A 3D object needs to be well unwrapped to easily manage texturing the images.



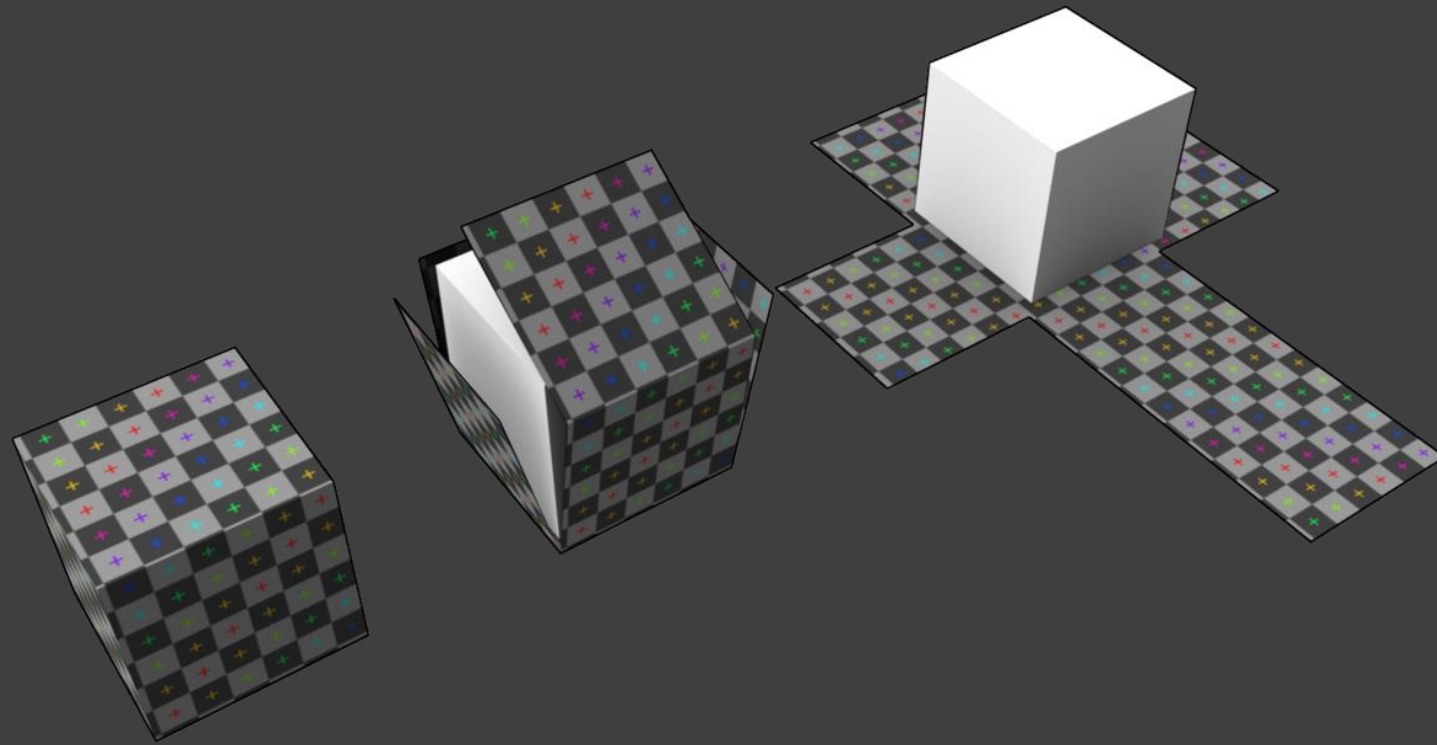
TEXTURING

Where does texture mapping take place?

- Texture mapping are implemented at the end of the rendering pipeline
- Remember the graphics pipeline, discussed in in chapter 4?



UV MAPPING & UNWRAPPING



UV MAPPING & UNWRAPPING

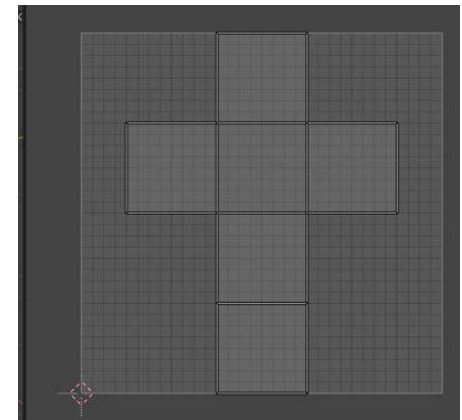
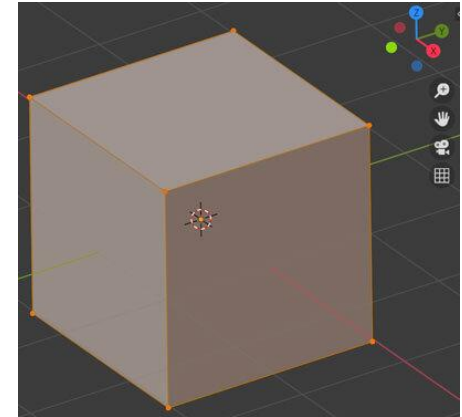
What is UV Mapping & Unwrapping?

- A UV map is the flat representation of the surface of a 3D model used to easily wrap textures. The process of creating a UV map is called UV unwrapping.
- The U and V refer to the horizontal and vertical axes of the 2D space, as X, Y and Z are already being used in the 3D space.
- Once the polygonal mesh has been created the next step is to unwrap/unfold it into a UV map.
- There is no such thing as a 3D texture, as they're always based on a 2D image.
- This is where UV mapping comes in, as it is the process of translating your 3D mesh into 2D information so that a 2D texture can be wrapped around it.

UV MAPPING & UNWRAPPING

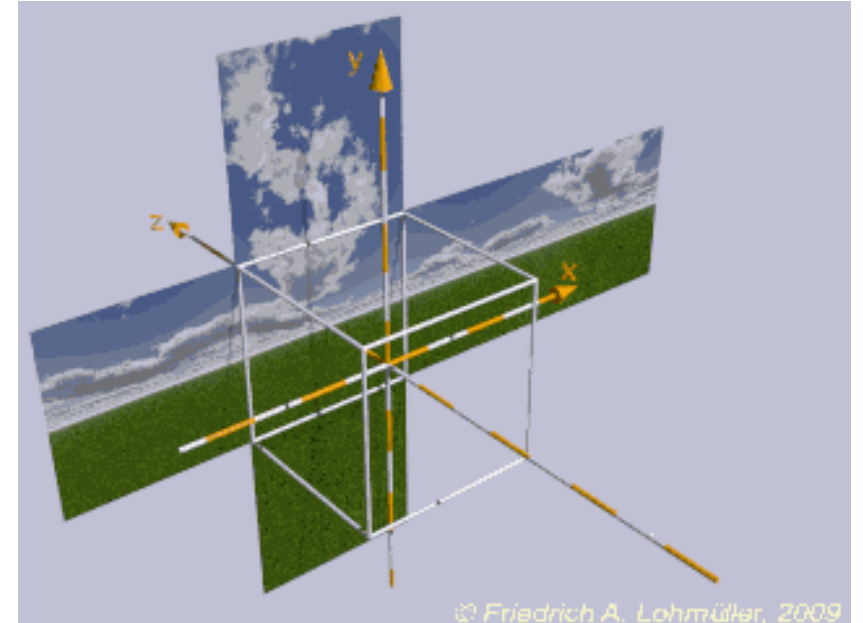
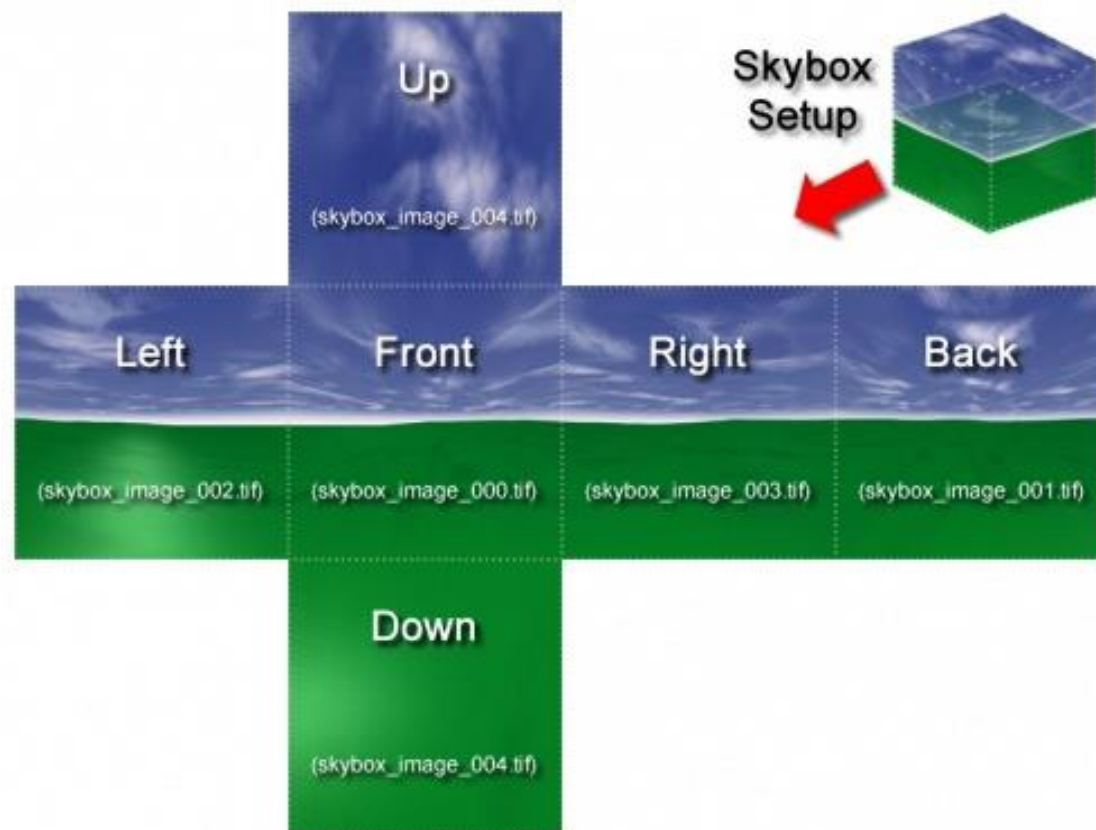
Unwrapping

- In a 2D object you do not need to perform the unwrapping process because a texturing image (which is 2D) easily can be mapped to a 2D object.
- A 3D object has faces in all three axes/all directions (top, bottom, left, right, front and back).
- Think of a cube! A cube is a basic example where the 3D application easily can unwrap it as seen in the figure.
- All the 6 faces are displayed on a 2D surface where you can easily manage the texturing.



UV MAPPING & UNWRAPPING

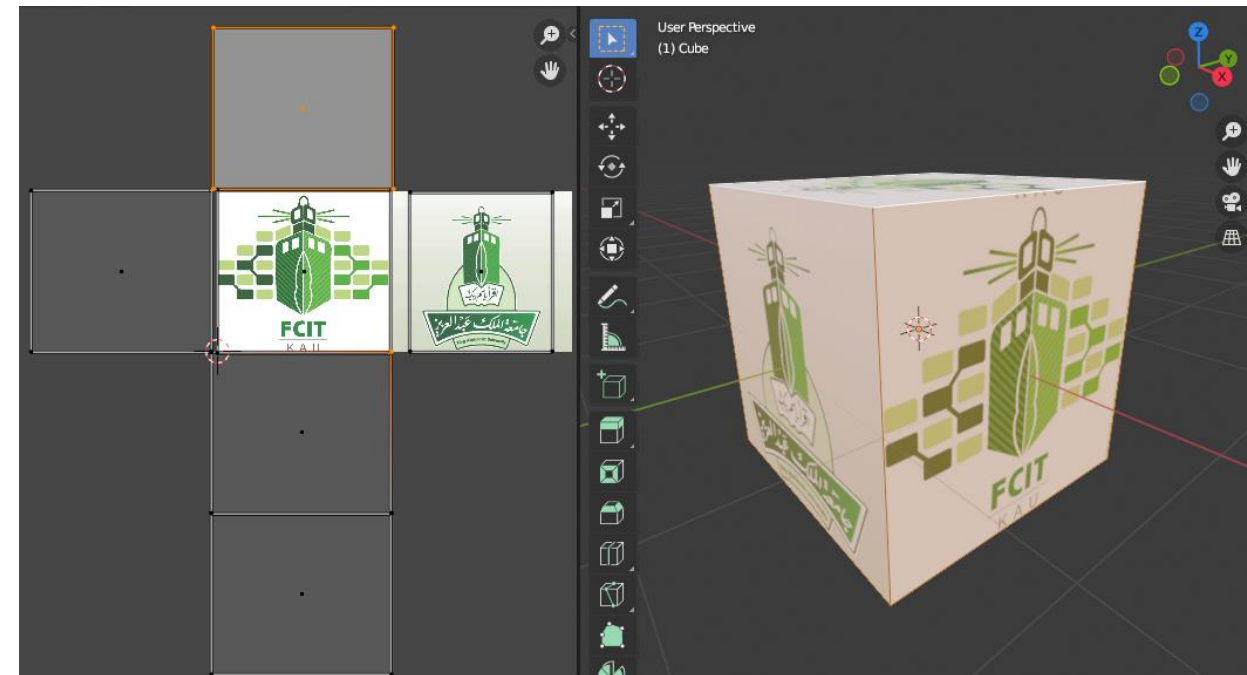
Unwrapping



UV MAPPING & UNWRAPPING

Unwrapping

- As you can see here in this example, the texture image has FCIT & KAU logos.
- On the front face of the cube, the KAU logo is textured and on the right face FCIT logo is textured.
- All the faces of the cube are unwrapped as appear on the left side and this gives you flexibility moving/scale/rotate faces of your 3D object for texture mapping (UV mapping).



UV MAPPING & UNWRAPPING

Unwrapping

- A cube is a basic example and as the meshes get more complex so do the UV maps. It can become quite a tedious process but it is essential to the 3D workflow.
- Think of this human model. His dress (Saudi Thoub) needs to be textured.
- There is no clear/well-organized unwrapping for the dress.
- It is your job to guide your 3D software where to unwrap.



UV MAPPING & UNWRAPPING

Unwrapping

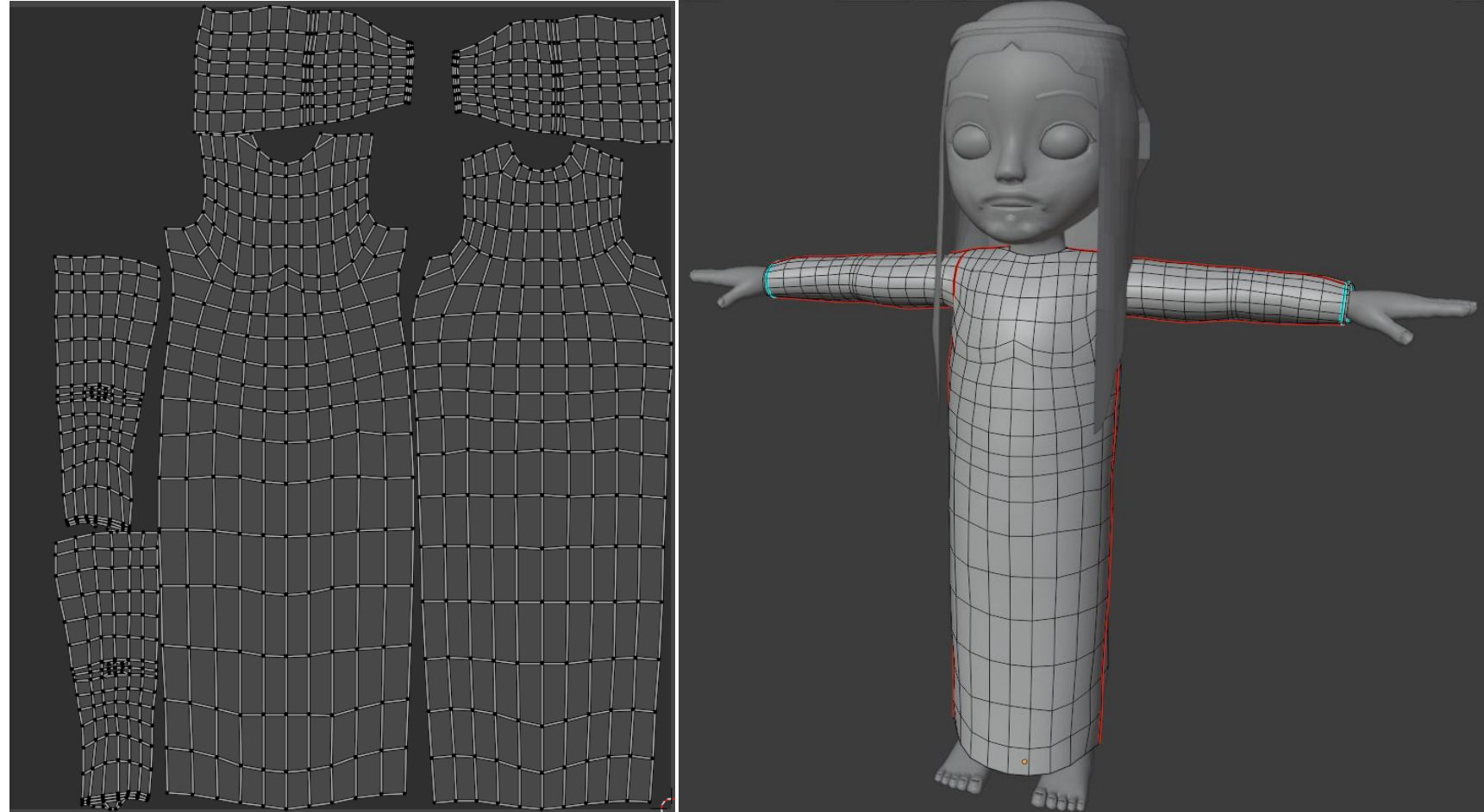
- Directing your 3D software on how to unwrap is by marking the edges where the cutting needs to be performed. The red lines appear in the figure.
- This process is called **seams**.
- Seams are an unfortunate and unavoidable side effect of flattening any 3D geometry.
- A seam is a part of the mesh that had to be split to be able to convert the 3D mesh into a 2D UV map.



UV MAPPING & UNWRAPPING

Unwrapping

- This is how the dress will be unwrapped based on the marked seams.
- With this well-organized unwrapping, clearly you can identify each part of the dress.
 - Your unwrapping may be different! It depends on where seams are marked.



UV MAPPING & UNWRAPPING

UV Mapping

- Once your 3D object is unwrapped in a well-organized way, it is the next step to map which parts of the textured image appear on which parts/ faces of your 3D object.
- Again, UV is the texture spaces. The texture space is assigned UV coordinates to be distinguished from XYZ for object coordinates.

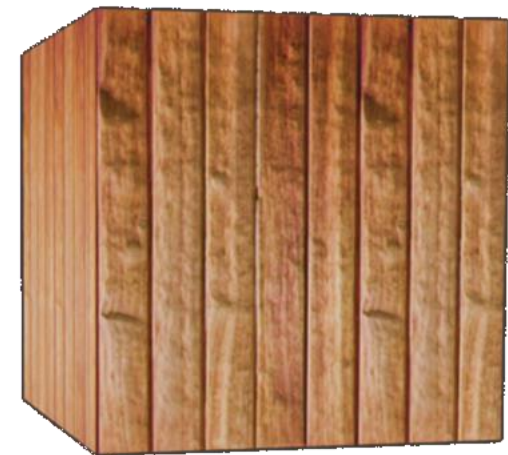


PHOTOREALISTIC TEXTURING



PHOTOREALISTIC TEXTURING

- What do you think of these two cubes simulating wooden boxes?
- Both are textured with one textured image, but each used different material techniques.
- Which of them gives a more realistic looking of wood?



PHOTOREALISTIC TEXTURING

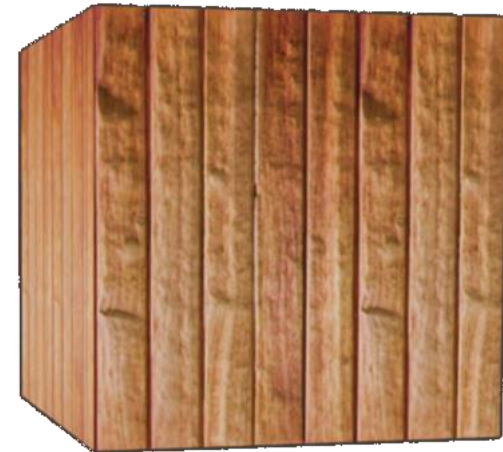
Basic Material



Texture image

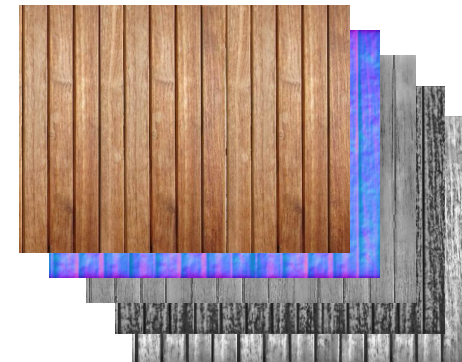


PBR Material



More Realistic Texturing!

Several
Texture Maps

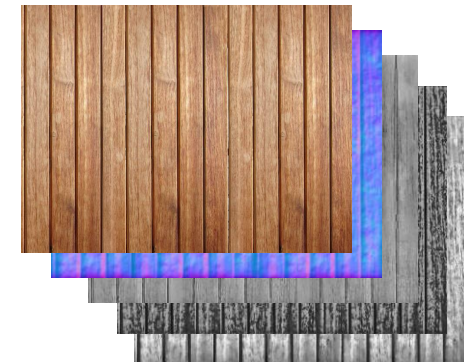
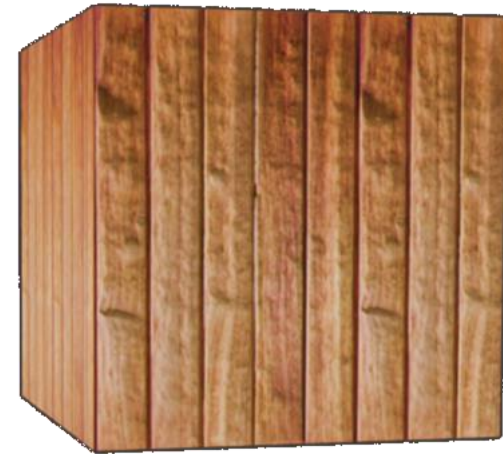


PHOTOREALISTIC TEXTURING

What is PBR Material?

- PBR stands for Physically Based Rendering
- It is used to develop for rendering incredibly photorealistic textures.
- The material describes the visual properties of a surface in a physically plausible way, such that realistic results are possible under all lighting conditions.
- It consists of several texture maps generated from the original texture image.

More Realistic Texturing!



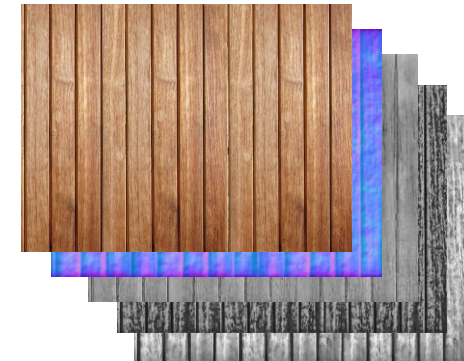
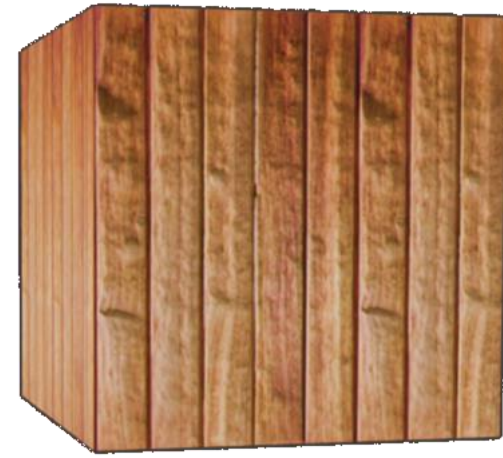
Several
Texture Maps

PHOTOREALISTIC TEXTURING

What is PBR Material?

- PBR Texture Maps:
 - Base color/Albedo map/Diffuse Map
 - Height Map
 - Normal Map
 - Roughness/Smoothness Map
 - Metalness Map
 - Ambient Occlusion/AO Map
 - Opacity/Transparency

More Realistic Texturing!



Several
Texture Maps

PHOTOREALISTIC TEXTURING

What is PBR Material?

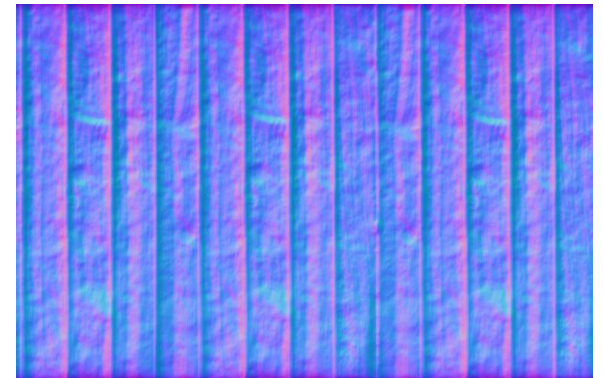
○ PBR Texture Maps:

- Base color/Albedo map/Diffuse Map

Simply, it's the original color or image you used for texturing. When using this map alone, it produces basic material for texturing.

- Normal Map

Normal maps are important to give your textures depth. This uses complex calculations to fake the way light interacts with the surface of the material to fake smaller bumps and dents. The base color of a normal map is a light purple, this is the “bottom” of the normal map that represents the surface of your polygonal mesh. From there RGB values are used to produce cracks, bumps, or pores in your model.



PHOTOREALISTIC TEXTURING

What is PBR Material?

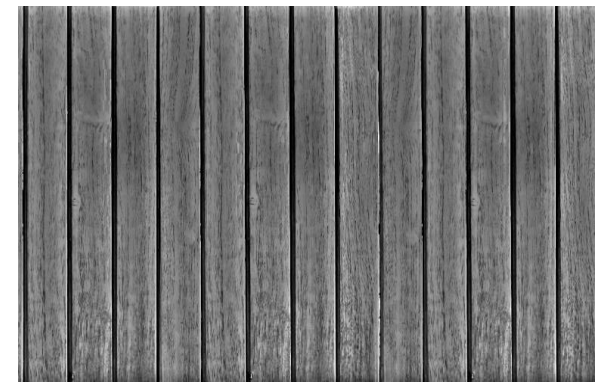
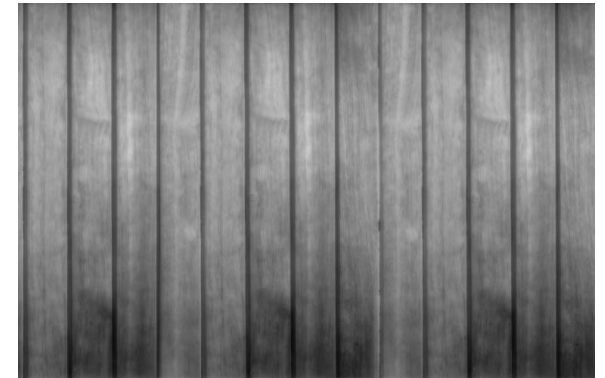
○ PBR Texture Maps:

■ Height Map

It is almost the same as normal maps to give the illusion of depth. Some applications use the Height map to generate the normal map. Wight color on the map makes the parts look upper whereas black makes the parts look lower.

■ Roughness/Smoothness/Glossiness Map

It defines how rough or smooth the surface is. Rough surfaces scatter the light in more directions than smooth surfaces, which makes reflections blurry rather than sharp. The value range is from 0.0 to 1.0. A value of zero where the object does not scatter light at all, making the reflections and lighting much sharper and brighter on the material.



PHOTOREALISTIC TEXTURING

What is PBR Material?

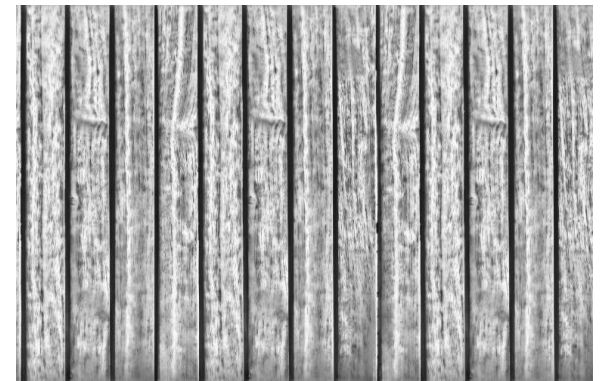
○ PBR Texture Maps:

■ Metalness Map

It affects how much a surface reflects the surrounding environment. Values range from 0.0 to 1.0. When metalness is 0.0, the base color is fully visible and the material looks like plastic or ceramics. When metalness is 1.0, the surface almost completely loses its base color and only reflects the surroundings.

■ Ambient Occlusion/AO Map

It is a map that is combined with the base color at render time to define how it reacts to light. It's a grayscale map with white being an area that will pick up the lightest with darker areas being more in shadow and less reactive to light.



PHOTOREALISTIC TEXTURING

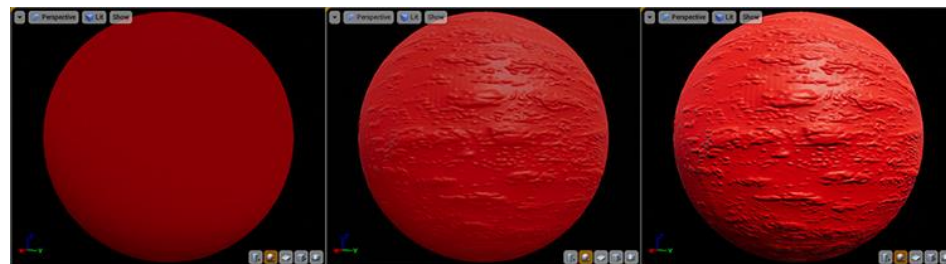
What is PBR Material?



Albedo 0%

Albedo 50%

Albedo 100%

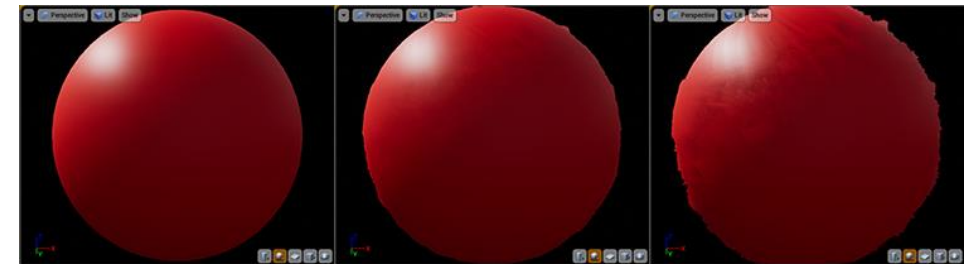


Normal 0%

Normal 50%

Normal 100%

<https://conceptartempire.com/texture-maps>



Height 0%

Height 50%

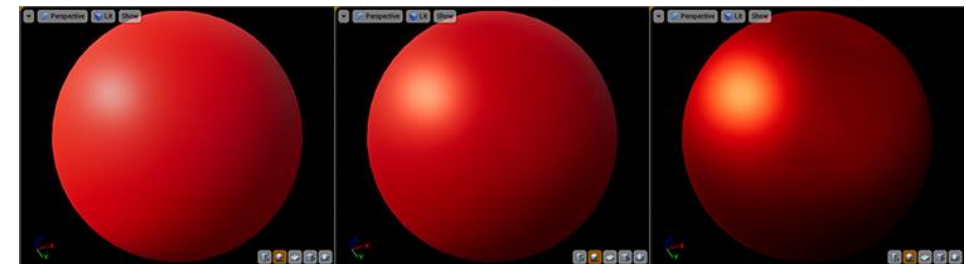
Height 100%



Roughness 0%

Roughness 50%

Roughness 100%



Metalness 0%

Metalness 50%

Metalness 100%

PHOTOREALISTIC TEXTURING

How to generate PBR maps?

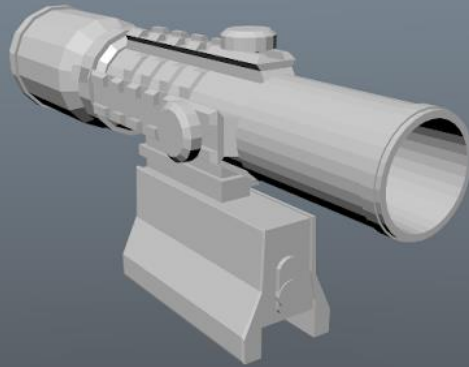
- Professionals share their generated PBR maps on texturing platforms such as www.textures.com
- Not always you can find your desired texturing available or for free.
- You can generate your own maps for a textured image using Photoshop. However, this requires a good level of dealing with photoshop.
- Online generators are available (some with less flexibility), such as [Normal Map Online](#)
- Desktop software is available. A good open-source one is [Materialize](#)
 - A tutorial on [how to use Materialize](#)

High Poly



373 414 Polygons

Low Poly



1542 Polygons

Low Poly + Normal Map



1542 Polygons

BAKING

—

BAKING

What is baking in graphics?

- Seems you are hungry! Thinking of a cack or biscuit?
- Baking in computer graphics is not something to eat 😊
- Baking is the name of the process of saving information related to a 3D mesh into a texture file (bitmap).
- Most of the time this process involves another mesh. In this case, the information of the first mesh is transferred onto the second mesh UVs and then saved into a texture.



BAKING

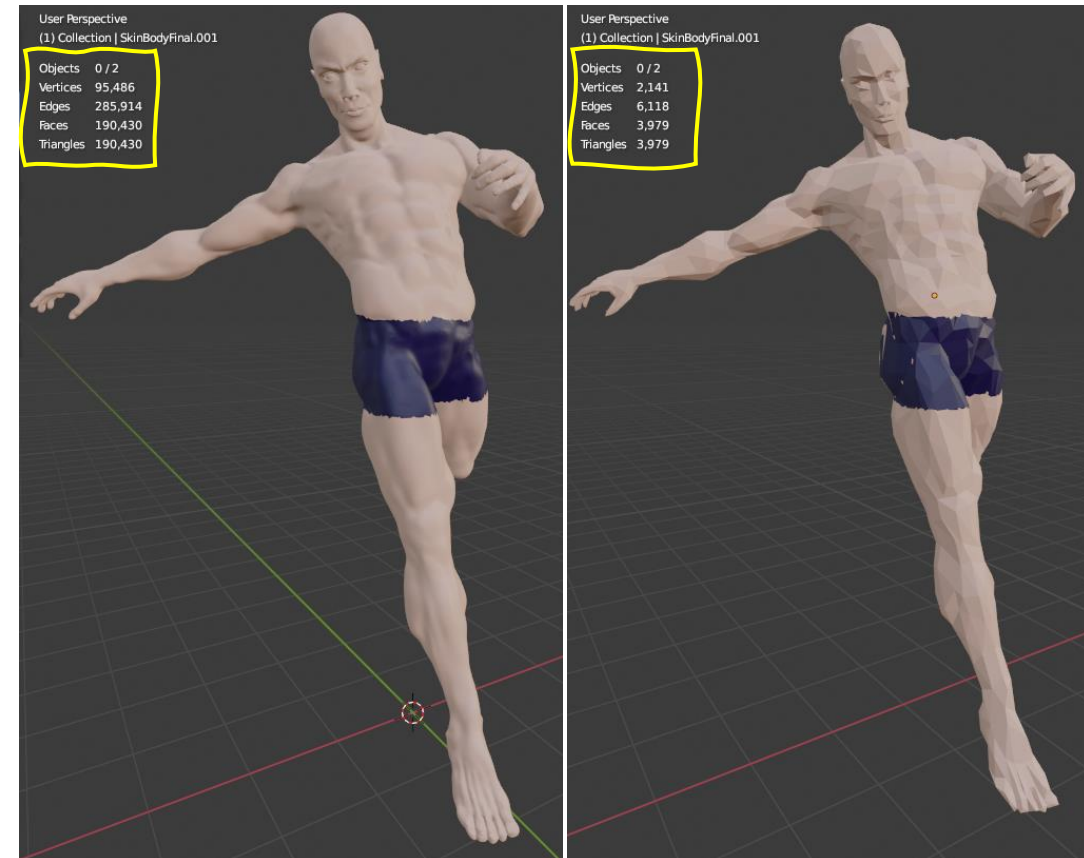
What is baking in graphics?

- Baking involves taking the details of a high poly model and texturing it to a duplicated/generated same model in a low poly resolution.
- Why generate a low poly model from a High poly?
 - To make it game ready assets
 - In the game, lots of animation processes. Animation involves thousands of calculations each second to keep drawing polygons.
 - Lots of polygons (3D primitives) in a model slow down the process...
 - Therefore, it is a wise decision to generate low poly models and save high poly details in a texture image.

BAKING

Why model a high poly model in the first place?

- Simple answer, to produce more realistic scenes that include details...
- It is much easier to have freedom modeling your objects in detail without worrying about a number of polygons.
- Once you are finished modeling a high poly, simply you can remodel/decimate a number of polygons to generate a low poly model.



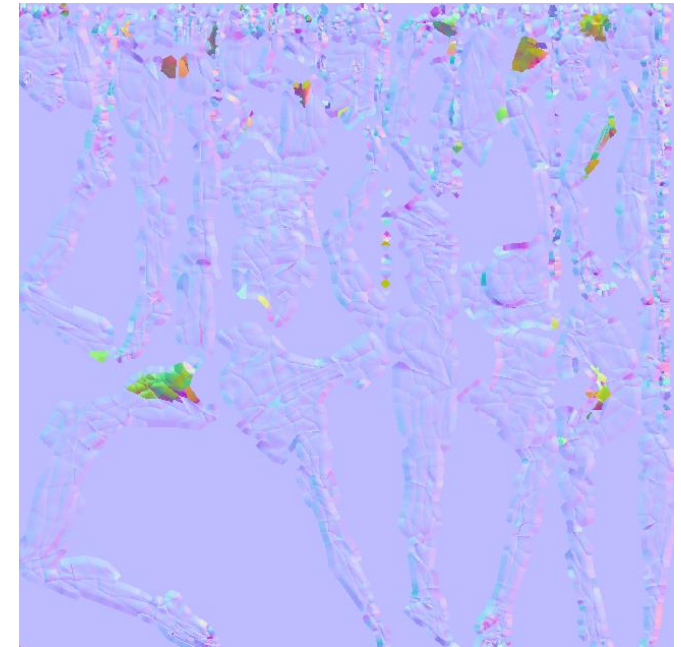
BAKING

How does baking work?

- PBR maps, in specific **Normal Map**, are generated from a high poly model.
- The generated Normal Map is used in texturing the low poly model to give the model the illusion of a high poly model without affecting the geometry.



The generated normal map of the high poly model

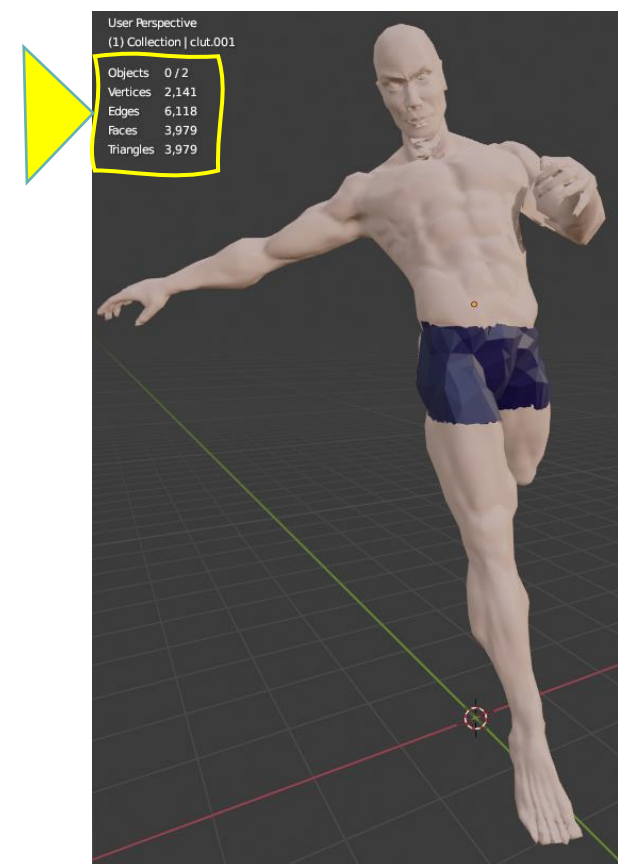
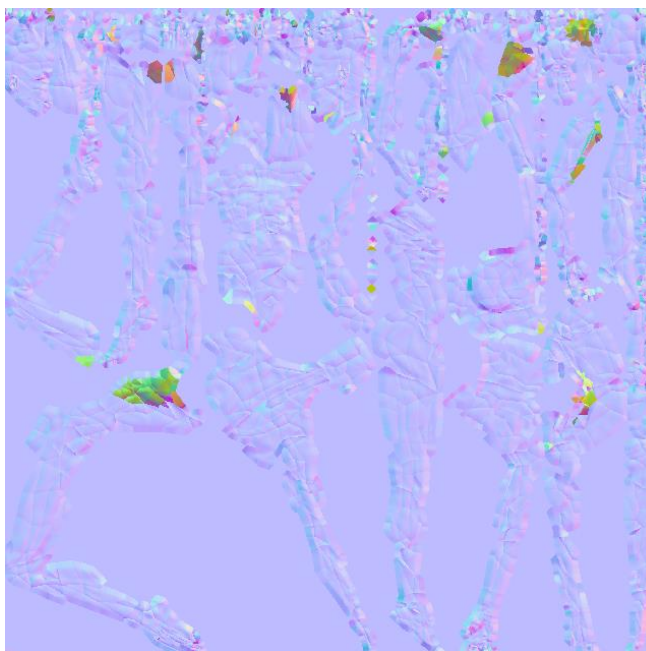


BAKING

How does baking work?



The generated normal map of the high poly model



The low poly model gets the detail of the HP model while geometry remained the same

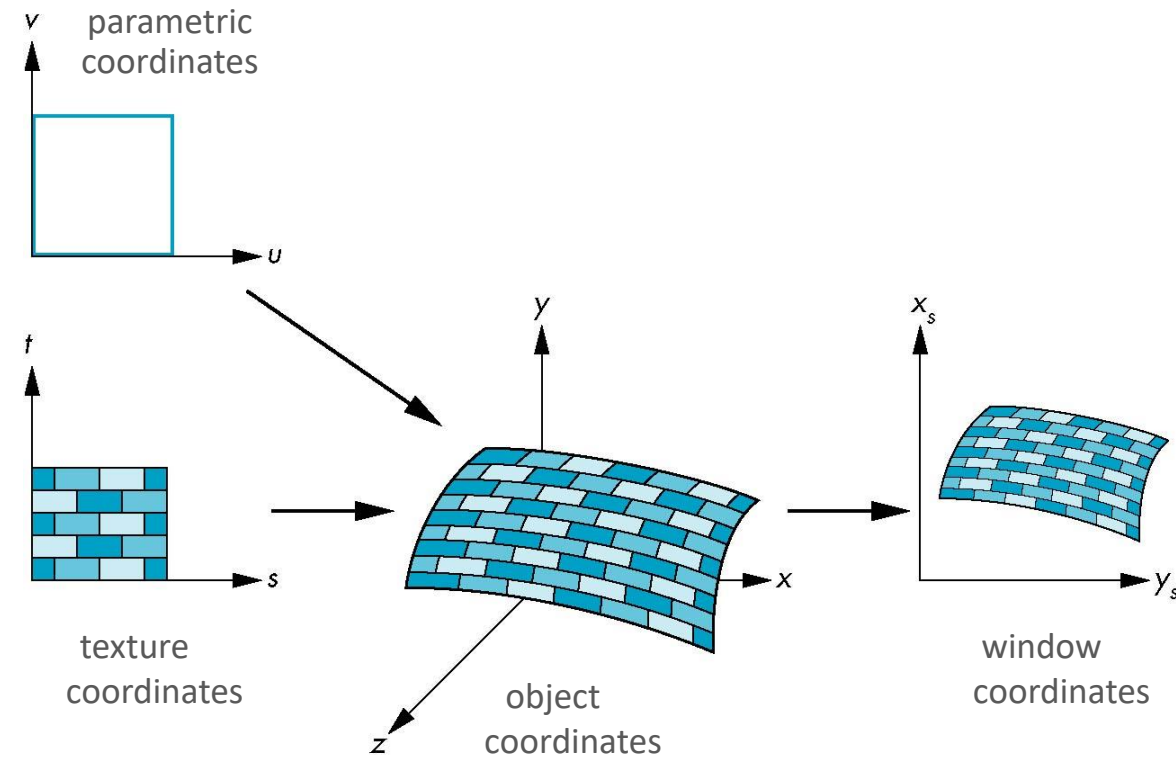
OPENGL TEXTURING FUNCTIONS



OPENGL TEXTURING FUNCTIONS

Coordinate Systems

- Parametric coordinates
 - Used for the model surfaces/faces
- Texture coordinates
 - Used to identify points in the image to be mapped
- Object or World Coordinates
 - Conceptually, where the mapping takes place (x, y, z)
- Window Coordinates
 - Where the final image is really produced



OPENGL TEXTURING FUNCTIONS

Coordinate Systems

- The basic techniques begin with some texture functions, `texture(s,t)` in texture space which is traditionally marked off by parameters named `s` and `t`.
- The function `texture(s,t)` produces a color or intensity value for each value of `s` and `t` between 0 and 1.
- The texture space is mapped to object space, `object(u,v)` using mapping transformations.
- Final mapping is then done from object space to image space, `image(x,y)` for display.
- `texture(s,t) → object(u,v) → image(x,y)`

OPENGL TEXTURING FUNCTIONS

Steps in Texture Mapping

To use texture mapping, you perform these steps.:

- Create a texture object and specify a texture for that object
- Specify texture properties and the texture image
- Enable texture mapping
- Provide the mapping between texture coordinates and the object's coordinates.

OPENGL TEXTURING FUNCTIONS

Steps in Texture Mapping

① Create a texture object and specify a texture for that object

- Each texture that OpenGL uses must be assigned a unique identifier.

```
glGenTextures(int, &texName);
```

- This is followed by calling `glBindTexture();` to specify what texture we are working with. Texture in OpenGL can be 1D or 2D, and even 3D exists. Normally we work with 2D and we specify it as follows

```
glBindTexture (GL_TEXTURE_2D, texName);
```

OPENGL TEXTURING FUNCTIONS

Steps in Texture Mapping

② Specify texture properties and the texture image

- When texture properties (parameters) are unidentified, default values are used.

Parameter	Values
GL_TEXTURE_WRAP_S	GL_CLAMP, GL_REPEAT
GL_TEXTURE_WRAP_T	GL_CLAMP, GL_REPEAT
GL_TEXTURE_MAG_FILTER	GL_NEAREST, GL_LINEAR
GL_TEXTURE_MIN_FILTER	GL_NEAREST, GL_LINEAR, GL_NEAREST_MIPMAP_NEAREST, GL_NEAREST_MIPMAP_LINEAR, GL_LINEAR_MIPMAP_NEAREST, GL_LINEAR_MIPMAP_LINEAR
GL_TEXTURE_BORDER_COLOR	any four values in [0.0, 1.0]

OPENGL TEXTURING FUNCTIONS

Steps in Texture Mapping

② Specify texture properties and the texture image

- Texture properties (parameters)

//The texture wraps over the edges (repeats) in the x direction

```
glTexParameteri(GL_TEXTURE_2D, GL_TEXTURE_WRAP_S, GL_REPEAT);
```

//The texture wraps over the edges (repeats in the y direction)

```
glTexParameteri(GL_TEXTURE_2D, GL_TEXTURE_WRAP_T, GL_REPEAT);
```

//when the texture area is large, repeat texel nearest center

```
glTexParameteri(GL_TEXTURE_2D, GL_TEXTURE_MAG_FILTER, GL_NEAREST);
```

//when the texture area is small, repeat texel nearest center

```
glTexParameteri(GL_TEXTURE_2D, GL_TEXTURE_MIN_FILTER, GL_NEAREST);
```

OPENGL TEXTURING FUNCTIONS

Steps in Texture Mapping

② Specify texture properties and the texture image

- Texture properties (parameters)



OPENGL TEXTURING FUNCTIONS

Steps in Texture Mapping

② Specify texture properties and the texture image

- Texture properties (parameters)



GL_REPEAT



GL_MIRRORED_REPEAT



GL_CLAMP_TO_EDGE



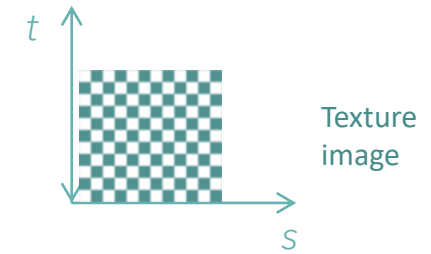
GL_CLAMP_TO_BORDER

OPENGL TEXTURING FUNCTIONS

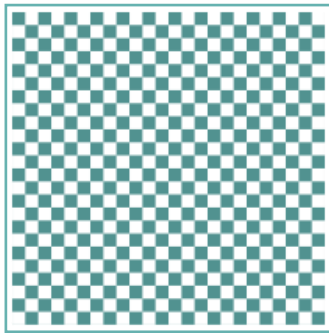
Steps in Texture Mapping

② Specify texture properties and the texture image

- Texture properties (parameters)



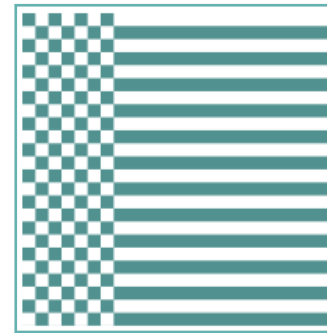
Wrap S : GL_REPEAT
Wrap T : GL_REPEAT



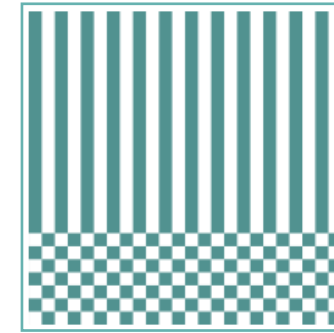
Wrap S : GL_CLAMP
Wrap T : GL_CLAMP



Wrap S : GL_CLAMP
Wrap T : GL_REPEAT



Wrap S : GL_REPEAT
Wrap T : GL_CLAMP



OPENGL TEXTURING FUNCTIONS

Steps in Texture Mapping

② Specify texture properties and the texture image

- Call `glTexImage*()`; to define the texture image used in each of texture object.

```
glTexImage2D(GL_TEXTURE_2D, 0, GL_RGBA, checkImageWidth,  
             checkImageHeight, 0, GL_RGBA, GL_UNSIGNED_BYTE, checkImage);
```

// border (no border)

//level (not multi-resolution)

OPENGL TEXTURING FUNCTIONS

Steps in Texture Mapping

③ Enable texture mapping

- Call `glEnable();` to turn on texture mapping
`glEnable (GL_TEXTURE_2D);`
- Call `glBindTexture();` again for the purpose that texture data becomes the current texture state
`glBindTexture(GL_TEXTURE_2D, texName);`

OPENGL TEXTURING FUNCTIONS

Steps in Texture Mapping

4 Mapping between the texture coordinates to the object coordinates

- Texture coordinates are often referred to as (s and t).
- For a texture image, s and t are in the range of 0 to 1.
- The mapping itself is similar to setting a color for each vertex.

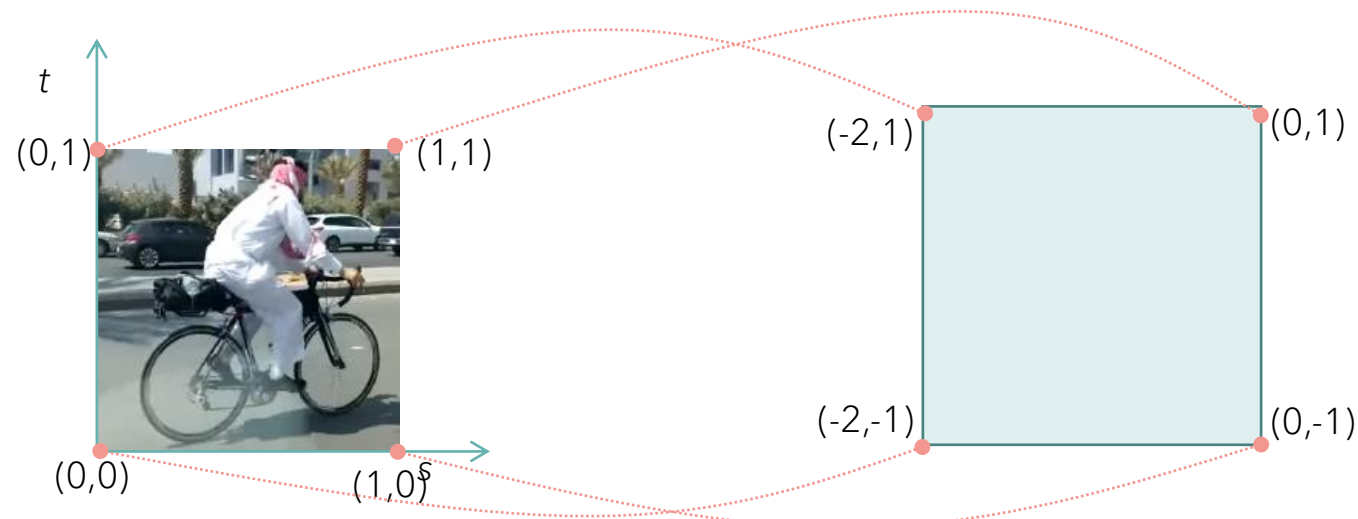
```
glBegin(GL_QUADS);  
    glTexCoord2f( 0.0, 0.0); glVertex3f(-2.0, -1.0, 0.0);  
    glTexCoord2f( 0.0, 1.0); glVertex3f(-2.0, 1.0, 0.0);  
    glTexCoord2f( 1.0, 1.0); glVertex3f( 0.0, 1.0, 0.0);  
    glTexCoord2f( 1.0, 0.0); glVertex3f( 0.0, -1.0, 0.0);  
glEnd();
```

OPENGL TEXTURING FUNCTIONS

Steps in Texture Mapping

4 Mapping between the texture coordinates to the object coordinates

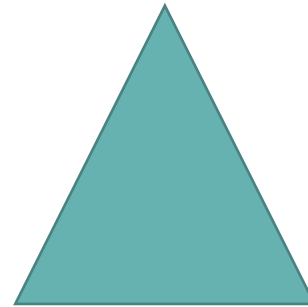
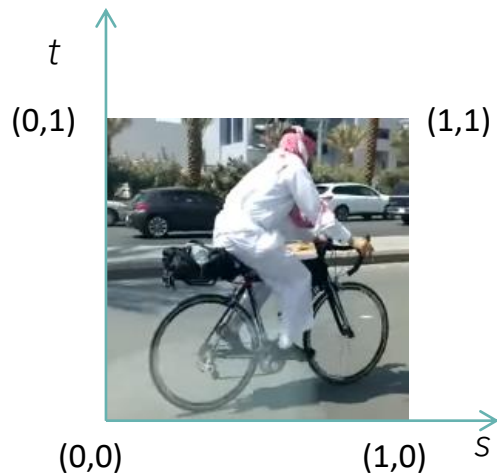
```
glTexCoord2f( 0.0, 0.0); glVertex3f(-2.0, -1.0, 0.0);  
glTexCoord2f( 0.0, 1.0); glVertex3f(-2.0, 1.0, 0.0);  
glTexCoord2f( 1.0, 1.0); glVertex3f( 0.0, 1.0, 0.0);  
glTexCoord2f( 1.0, 0.0); glVertex3f( 0.0, -1.0, 0.0);
```



OPENGL TEXTURING FUNCTIONS

Steps in Texture Mapping

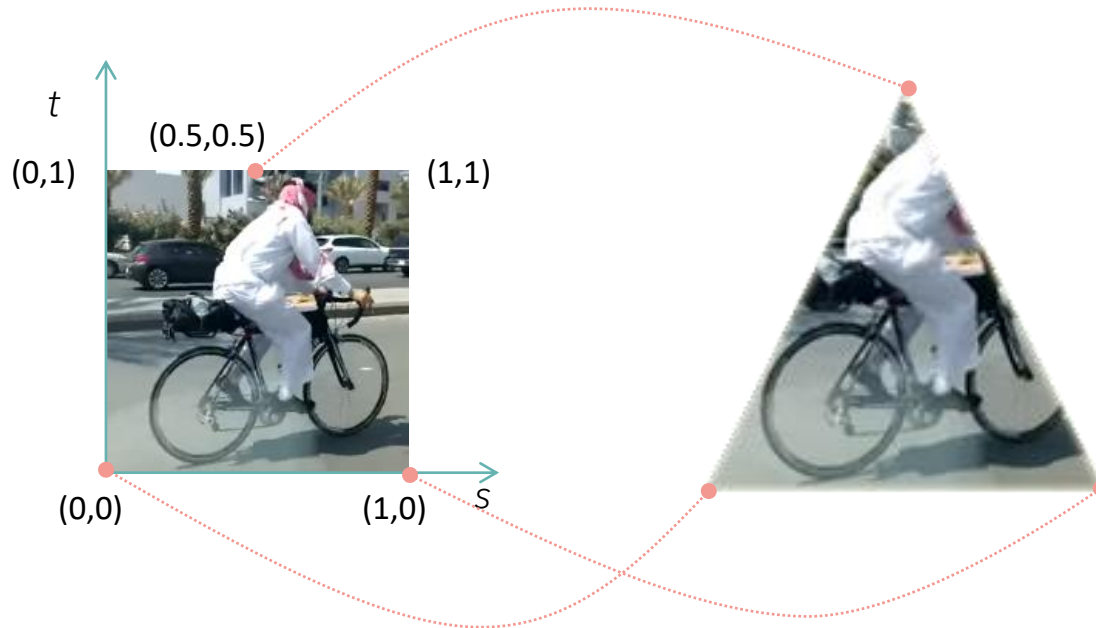
- 4 Mapping between the texture coordinates to the object coordinates



OPENGL TEXTURING FUNCTIONS

Steps in Texture Mapping

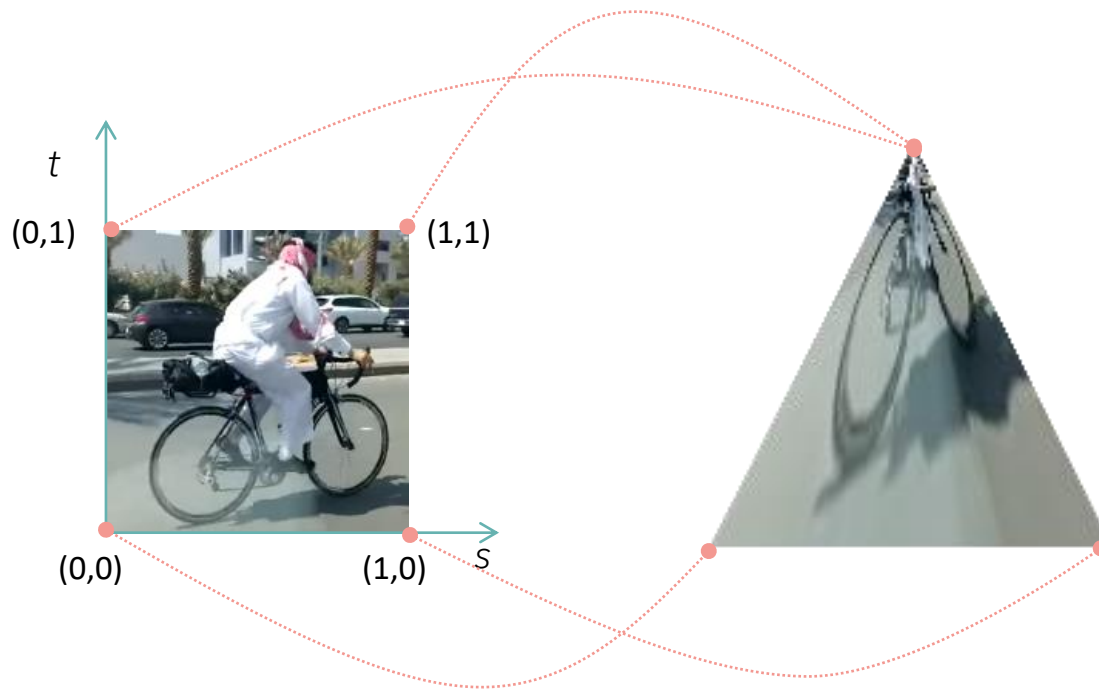
④ Mapping between the texture coordinates to the object coordinates



OPENGL TEXTURING FUNCTIONS

Steps in Texture Mapping

④ Mapping between the texture coordinates to the object coordinates



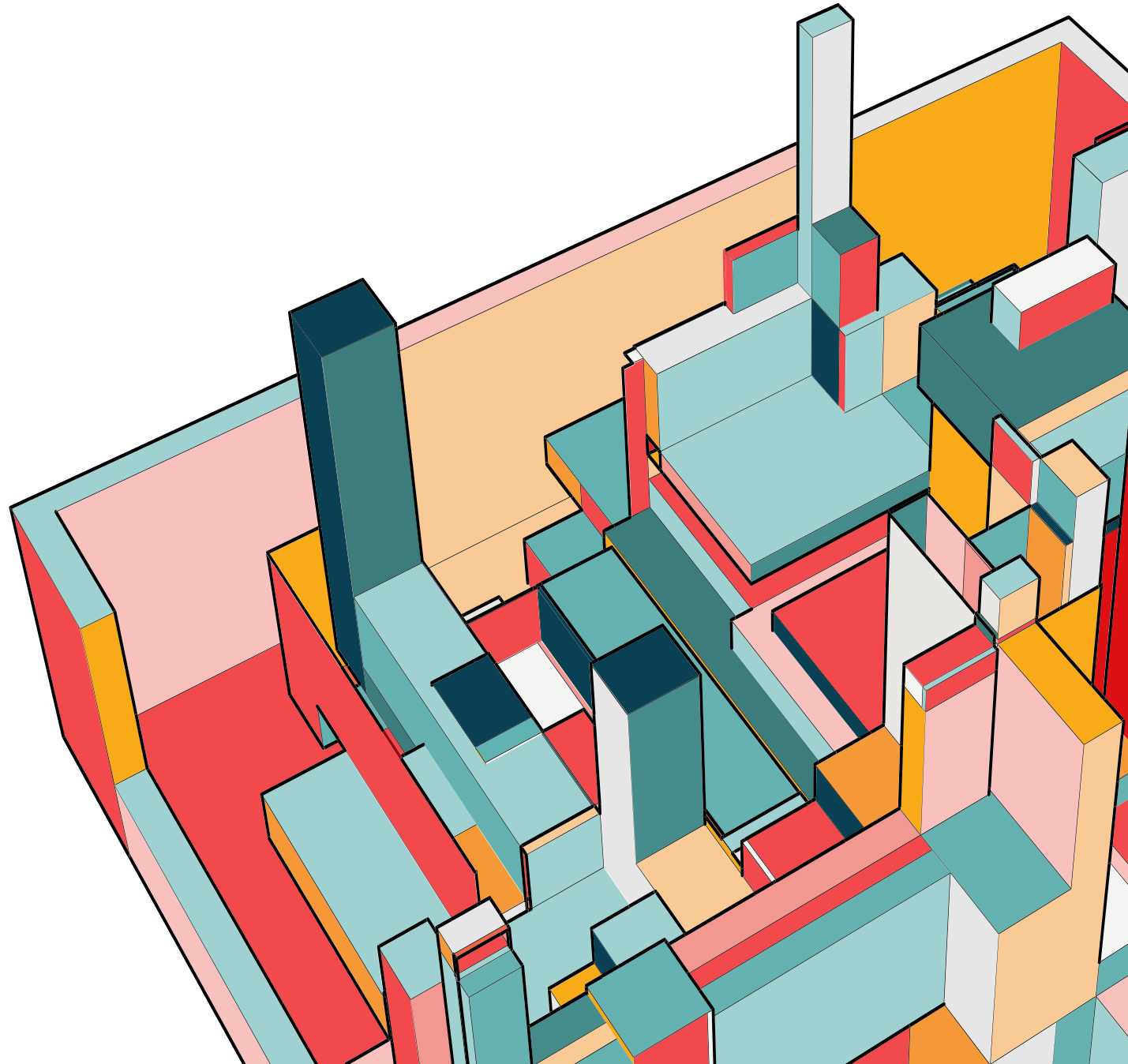
REFERENCES

- Concept Empire: Texture Maps, UV Unwrapping
- Microsoft: PBR materials
- Lee Stemkoski, 2021. Developing Graphics Frameworks with Python and OpenGL.
- Ed Angel, 2009. Interactive Computer Graphics 5E: OpenGL Texture Mapping.
- Light Collab: Cross-Polarization Photography for Textures

100% CONFIDENT

Your journey in this course is
enjoyable and beneficial!

Enjoy learning CPIT285!



THANK YOU

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